A conservative receiver and the origin of a correlation scale

A finite harmonic receiver has a vanishing long-window action observable at fixed centre velocity. An increasing periodic chain instead has plateau $\Theta\sqrt{m/k}$ when its size tends to infinity first, with mode energy Θ and spring stiffness k. A common hard particle-speed ceiling on the stated phase preparation forces this plateau to close. The comparison locates the positive scale in the low-frequency weights and energy preparation.

1. Hamiltonian and invariant preparation

Let $x, p \in \mathbb{R}^n$, $n \geq 2$, $M = \operatorname{diag}(m_1, \ldots, m_n)$ with $m_i > 0$, and let K be a connected spring-network Laplacian. Thus $K = K^T \geq 0$ and $\ker K = \operatorname{span}\{\mathbf{1}\}$. The Hamiltonian and equations are

$$\mathcal{H} = \tfrac{1}{2}p^T M^{-1} p + \tfrac{1}{2}x^T K x, \qquad \dot{x} = M^{-1}p, \quad \dot{p} = -Kx.$$

Energy and total momentum are conserved. These linear equations give global smooth trajectories. Write $D = M^{-1/2} K M^{-1/2}$ and choose an orthonormal eigenbasis $e_0, e_1, \ldots, e_{n-1}$, with $e_0 = M^{1/2}\mathbf{1}/\sqrt{M_{\mathrm{tot}}}$ and positive eigenvalues ω_j^2 for $j \geq 1$. Define canonical coordinates $q_j = e_j^T M^{1/2} x$, $P_j = e_j^T M^{-1/2} p$.

Fix centre velocity $V_0 = 0$ and centre position. Assign internal mode energies $E_j \geq 0$, and independent phases ϕ_j uniform on $[0, 2\pi)$. Then

$$q_j(t) = \frac{\sqrt{2E_j}}{\omega_j}\cos(\omega_j t + \phi_j), \qquad P_j(t) = -\sqrt{2E_j}\sin(\omega_j t + \phi_j).$$

This is an invariant probability ensemble on the internal phase tori, even when frequencies are commensurate. It is a preparation, not a mixing premise. The total energy is $E = \sum_j E_j$, so $|\dot{x}_i| \leq \sqrt{2E/m_i}$ for every trajectory. Choosing $E < \frac{1}{2}(\min_i m_i)c^2$ puts all particles below c in this frame. These are Newtonian springs with a speed-bounded preparation; propagation of signals and Lorentz covariance are separate model requirements.

2. Exact action observable and its limits (C047)

For tagged particle i, let $b_{ij} = e_{ji}/\sqrt{m_i}$, where e_{ji} denotes component i of vector e_j. Then $v_i(t) = \sum_{j\geq 1} b_{ij} P_j(t)$ has zero mean and covariance

$$C_i(t) = \sum_{j\geq 1} w_{ij}\cos(\omega_j t), \qquad w_{ij} = b_{ij}^2 E_j \geq 0.$$

Phase averaging gives this identity: distinct modes have zero cross covariance, and one phase has $\mathbb{E}[P_j(t)P_j(0)] = E_j\cos(\omega_j t)$. Integrating the covariance over the displacement square yields, for $\Delta > 0$,

$$\mathsf{h}_i(\Delta) = \frac{m_i}{\Delta}\operatorname{Var}[x_i(t+\Delta) - x_i(t)] = \frac{2m_i}{\Delta}\sum_{j\geq 1}\frac{w_{ij}}{\omega_j^2}[1 - \cos(\omega_j\Delta)].$$

The observable has action units and is independent of preparation time t. Set $B_i = \sum_j w_{ij}/\omega_j^2$, with units of length squared. Then

$$0 \leq \mathsf{h}_i(\Delta) \leq \frac{4m_i B_i}{\Delta} \longrightarrow 0 \quad (\Delta \to \infty).$$

At short windows, $\mathsf{h}_i(\Delta) = m_i C_i(0)\Delta + O(\Delta^3)$. The finite cosine correlation has persistent memory; its ordinary improper Green–Kubo integral need not exist. The finite-window formula above supplies the relevant limit directly, without replacing the Hamiltonian dynamics by a reversible dissipative generator.

3. Centre motion and the two-body receiver (C047)

An independent random constant centre velocity with variance σ_0^2 adds $m_i\sigma_0^2\Delta$ to h_i. This follows from $x_i(t+\Delta)-x_i(t) = V_0\Delta+$ internal displacement. The velocity ensemble is stationary; for nonzero centre motion the absolute-position ensemble need not be. A deterministic centre velocity contributes zero to the variance and can be removed by a change of frame.

For two particles of masses m, M_r coupled by a spring $k > 0$, put $\mu = mM_r/(m + M_r)$ and $\omega^2 = k/\mu$. At zero total momentum, $x_1 = R + [M_r/(m + M_r)]r$, and assign relative energy E with uniform phase. The tagged response is

$$\mathsf{h}_1(\Delta) = \frac{2mE}{k\Delta}\left(\frac{M_r}{m + M_r}\right)^2 [1 - \cos(\omega\Delta)].$$

The spring is a genuine momentum receiver: $m\ddot{x}_1 = -kr$ and $M_r\ddot{x}_2 = kr$. Its reversible exchange produces the oscillatory response rather than the exponential memory law supplied by A02's refreshed bath. Scaling every E_j by a^2, $0 < a \le 1$, scales the entire response by a^2 while preserving the equations, conservation laws and speed margin.

4. What the large-receiver test must change (C047)

For a sequence of receivers and windows $\Delta_N \to \infty$, the bound $\mathsf{h}_{i,N}(\Delta_N) \le 4m_{i,N}B_{i,N}/\Delta_N$ shows that a positive limiting response requires this upper bound to stay positive. In particular, uniform control of $m_{i,N}B_{i,N}$ forces a zero limit. Soft modes can enlarge $B_{i,N}$, but their weights depend on the measured coordinate and the preparation. This is the next selection test.

The finite model yields no physical-time attraction from the invariant preparation: the observable already has no t dependence. Next specify an increasing spring network, its energy allocation and tagged spectral measure, then compare taking receiver size and observation duration to infinity in opposite orders. Derive any surviving constant from these mechanical inputs. Mesh refinement of the same smooth trajectories is a third, distinct limit.

5. An increasing periodic chain (C048)

Take odd $N \ge 3$ equal masses m on a ring, with spring energy $\frac{k}{2}\sum_{i=0}^{N-1}(x_{i+1} - x_i)^2$ (indices modulo N). Fix the centre position and momentum. Give every real internal normal mode energy $\Theta > 0$ and independent uniform phase. Here Θ is a preparation energy, and total energy is $(N - 1)\Theta$. The mode frequencies and tagged covariance are

$$\omega_j = \Omega\sin(\pi j/N), \quad \Omega = 2\sqrt{k/m}, \qquad C_N(t) = \frac{\Theta}{mN}\sum_{j=1}^{N-1}\cos(\omega_j t).$$

The real sine/cosine eigenvectors have equal mode energies, so their paired squared components sum to $2/N$ at each site. Applying §2 gives

$$h_N(\Delta) = \frac{2\Theta}{N\Delta}\sum_{j=1}^{N-1}\frac{1 - \cos(\omega_j\Delta)}{\omega_j^2}.$$

For each fixed Δ, this Riemann sum converges to

$$h_\infty(\Delta) = \frac{2m}{\Delta}\int_0^\Omega \frac{1 - \cos(\omega\Delta)}{\omega^2}\rho(\omega)\, d\omega, \quad \rho(\omega) = \frac{2\Theta}{\pi m\sqrt{\Omega^2 - \omega^2}}.$$

This is a statement about limits of tagged covariances and increment variances; it does not require a stationary absolute-position law for the infinite chain. The density is integrable at its upper edge and continuous at zero. Splitting at any fixed $0 < a < \Omega$, the contribution from $[a,\Omega]$ is $O(1/\Delta)$. In the lower interval substitute $y=\omega\Delta$. Dominated convergence, using $(1-\cos y)/y^2\in L^1(0,\infty)$ and bounded ρ on $[0,a]$, gives

$$\lim_{\Delta\to\infty} h_\infty(\Delta)=\pi m\rho(0)=\frac{2\Theta}{\Omega}=\Theta\sqrt{m/k}.$$

The integral $\int_0^\infty(1-\cos y)y^{-2}dy=\pi/2$ follows by integration by parts and the Dirichlet sine integral. Thus

$$\lim_{\Delta\to\infty}\lim_{N\to\infty} h_N(\Delta)=2\Theta/\Omega, \qquad \lim_{N\to\infty}\lim_{\Delta\to\infty} h_N(\Delta)=0.$$

A joint regime is also explicit. Write $f(z)=2(1-\cos z)/z^2=2\int_0^1(1-r)\cos(zr)\,dr$ with $f(0)=1$. Then $|f'|\le 1/3$. The integrand $\Theta\Delta f(\Omega\Delta\sin\pi x)$ is Lipschitz with constant at most $\pi\Theta\Omega\Delta^2/3$. A left Riemann sum error is at most that constant divided by $2N$; removing its zero mode contributes $\Theta\Delta/N$. Therefore

$$|h_N(\Delta)-h_\infty(\Delta)|\le\frac{\pi\Theta\Omega\Delta^2}{6N}+\frac{\Theta\Delta}{N}.$$

At fixed m,k,Θ, $\Omega\Delta_N\to\infty$ and $(\Omega\Delta_N)^2/N\to 0$ give the positive joint limit $2\Theta/\Omega$. The action scale is selected by the input energy and spring timescale.

6. Energy and a common speed ceiling (C049)

The positive-plateau preparation has extensive total energy. Its phase support also contains increasingly large tagged velocities. At site zero use the standard real Fourier basis. Only the $(N-1)/2$ cosine modes contribute; each has maximal velocity contribution $2\sqrt{\Theta/(mN)}$. Phases can align, so the exact support maximum is

$$\sup_\phi |v_0|=(N-1)\sqrt{\Theta/(mN)}.$$

Every neighbourhood of the maximizing phases has positive preparation measure. A common ceiling $|v_i|\le c$ for all supported trajectories thus requires $\Theta_N\le mc^2N/(N-1)^2=O(1/N)$. This is a necessary condition from one site; no sufficiency for all sites is inferred from that maximum.

Both bounded total energy and this necessary speed condition close the response uniformly in the window. Indeed, $1-\cos z\le z^2/2$ gives $h_N(\Delta)\le\Theta_N\Delta$. The identity $\sum_{j=1}^{N-1}\csc^2(\pi j/N)=(N^2-1)/3$ gives

$$h_N(\Delta)\le\frac{4\Theta_N(N^2-1)}{3N\Omega^2\Delta}, \qquad \sup_{\Delta>0} h_N(\Delta)\le\frac{2\Theta_N}{\Omega}\sqrt{\frac{N^2-1}{3N}}.$$

The trigonometric identity follows by differentiating $\sum_{j=0}^{N-1}\cot(x+\pi j/N)=N\cot(Nx)$ and subtracting the $j=0$ singularity as $x\to 0$. The last inequality uses $\min(A\Delta,B/\Delta)\le\sqrt{AB}$. For $\Theta_N=O(1/N)$ the uniform bound tends to zero. Thus this explicit positive reservoir plateau uses a different preparation class from a family with a common strict particle-speed ceiling. Its dependence on Θ also leaves preparation-independent action selection open.

7. Source input and reproduction

Ford–Kac–Mazur, printed p. 505, (2)–(5), supplies the matrix harmonic propagator. Their canonical preparation excludes zero eigenvalues at that stage; ours fixes the centre and uses compact fixed-energy phase tori. Zwanzig, p. 219, (21)–(24), supplies a complementary reduced-description route: eliminate harmonic bath coordinates to obtain a cosine memory kernel. That friction kernel differs from the tagged velocity covariance calculated here. Its continuum replacement must be propagated through the tracer equation before identifying a displacement coefficient.

B23 records the established ingredients, derived finite-network consequence and bounded reading coverage. Historical modal, two-body and three-body checks accompany C047's proof. C047 has coordinator proof review and a sequential Luna-low prior-art audit. Its limits concern the stated observable and preparation class.

B24 audits §§5–6 against Ford–Kac–Mazur's cyclic Fourier construction and its finite-sum/infinite-chain distinction, printed p. 506, (12)–(24). The tagged density, plateau and speed-support bounds are derived here with fixed-energy phases. Six exact identities, four finite ring spectrum/cosecant comparisons and four Riemann-bound tests were recorded before the hard no-Python-verification rule; the handoff records that history. The new uncommitted verification script and output were withdrawn under the rule. Current mathematical verification uses the written proofs and source review. The next mechanism must retain bounded velocities while supplying low-frequency response; its preparation dependence remains a separate selection test.

8. What information survives a cut?

The fixed-energy two-body receiver gives an exact cut-state countertest: position-only stationary conditional kernels fail composition, while retaining momentum restores it. Independently refreshing direction at every inserted cut preserves one-time position marginals but freezes terminal position as the mesh shrinks. These are C062–C063; the full maps and proof are in the cut-state note, with the B33 audit.

With $q = x - X$, $\mu = mM/(m + M)$, $\omega = \sqrt{k/\mu}$ and $A = \sqrt{2E/k}$, uniform phase on the energy ellipse gives the conditional one-segment positions

$$q' = q\cos\omega t \ \pm \sqrt{A^2 - q^2}\sin\omega t$$

with equal weights. Thus the full mean after $s + t$ is $q\cos\omega(s + t)$, whereas independent conditional composition gives $q\cos\omega s\cos\omega t$. The missing interface variable is $p_s = \mu\dot{q}_s$ in $q_{s+t} = q_s\cos\omega t + p_s\sin\omega t/(\mu\omega)$. It has momentum units. Keeping this full internal state composes exactly; marginalizing the true joint path law also preserves all cut restrictions.

For N independent resets at intervals T/N, direct iteration gives

$$\mathbb{E}Q_N = q[\cos(\omega T/N)]^N,$$
$$\mathbb{E}Q_N^2 = \frac{A^2}{2} + \left(q^2 - \frac{A^2}{2}\right)[\cos(2\omega T/N)]^N.$$

The first limit is q and the second is q^2, proving terminal conditional L^2 convergence to the starting position. Under the original stationary preparation the reset two-time correlation tends to $A^2/2$, while the true value is $(A^2/2)\cos\omega T$. A cut that only observes the motion performs no reset and leaves the true correlation unchanged.

The normalized orbital action is $J = (2\pi)^{-1}\oint p\,dq = E/\omega$: integrating $\mu\omega A^2\sin^2\phi$ over a cycle proves the formula. Uniform-phase covariance also has square-root determinant E/ω. Cooling $E \downarrow 0$ at fixed Hamiltonian closes both, with exact phase-state composition intact. R04 extends this test to a third body, as follows.

9. A third body: exact memory and recoverable history

Retaining tagged momentum leaves receiver memory in the equal-mass open three-body chain. At zero centre position and momentum, let $x = x_1$, $y = x_2 - x_3$, $\mu = 3m/2$, $\nu = m/2$, $P = \mu\dot{x}$ and $Q = \nu\dot{y}$. Physical tagged momentum is $p_1 = 2P/3$. The internal Hamiltonian is

$$H = \frac{P^2}{2\mu} + \frac{Q^2}{2\nu} + \frac{9k}{8}x^2 - \frac{3k}{4}xy + \frac{5k}{8}y^2.$$

For fixed microcanonical energy $E > 0$, eliminate y, Q exactly. With $g = 3k/4$, $d = 5k/4$ and $\Omega^2 = 5k/(2m)$,

$$\mu\ddot{x} + \frac{9k}{5}x + \int_0^t \frac{9k}{20}\cos\Omega(t-s)\dot{x}(s)\,ds = F_0(t),$$

$$F_0(t) = g\left(y_0 - \frac{g}{d}x_0\right)\cos\Omega t + \frac{gQ_0}{\nu\Omega}\sin\Omega t.$$

These inherited quadratures carry the receiver information across a cut. The kernel has stiffness units; its convolution is a force. The full proof in R04 also recovers the hidden pair from tagged phase at two sufficiently close times. The relevant block determinant is $g^2\delta^4/(12\mu\nu) + O(\delta^6)$; inverse entries grow up to δ^{-3}. Exact recovery and finite-precision physical readout therefore pose different requirements.

The internal frequencies are $\sqrt{k/m}$ and $\sqrt{3k/m}$, and modal actions are E_j/ω_j. Cooling scales the actions to zero while keeping the memory kernel fixed. C064–C065 and B34 record this derived specialization of harmonic-bath elimination and observability. R05 supplies the following readout test.

10. Readout precision and a scalable classical instrument

Two sequential classical probes read the tagged pair with vanishing error and back-reaction under increasingly concentrated preparations. With canonical probe pairs (q, π), (r, ρ) and integrated interaction Hamiltonians $G_x = \alpha x\pi$, $G_P = \beta P\rho$, the pre-cut estimates and kicks are

$$\widehat{x} = x + q/\alpha, \qquad \widehat{P} = P + r/\beta - \alpha\pi,$$

$$d_x = \beta\rho, \qquad d_P = -\alpha\pi.$$

Here α is dimensionless and β has units T/M. For incoming rectangular support half-widths (s_q, t_q, s_r, t_r), the intrinsic support accuracy-disturbance products are $s_q t_q$ and $s_r t_r$, in action units. Both can approach zero across ordinary positive-volume classical preparations.

At fixed horizon T, put $\eta = (T/N)/t_*$ with fixed time unit t_*. Widths $O(\eta^4)$ yield hidden-state recovery error $O(\eta)$ after the cubic inverse amplification, including the intervening kick. The accumulated state disturbance is $O(\eta^3)$ uniformly over the horizon. The original receiver preparation stays fixed, while fresh probe count and total fixed-mass probe mass grow as $O(\eta^{-1})$.

C066–C067, the full readout proof and B35 specify this externally switched impulsive instrument. Its ideal position records and shrinking probe widths leave preparation and recording resources open. R06 supplies the finite-duration construction below; readout conditioning and canonical reaction alone have not excluded zero action.

11. Finite-duration readout with an autonomous clock

One mechanical clock and four free probes suffice for delayed reconstruction of the three-body receiver at fixed total apparatus mass. Let (s, p_s) be the clock pair, (q_j, π_j) the probe pairs, and all five masses be positive and fixed. With fixed stiffness K and dimensionless $0 < \lambda \le \lambda_0$, the autonomous Hamiltonian is

$$H_\lambda = H + \frac{p_s^2}{2M_c} + \sum_{j=1}^{4} \frac{\pi_j^2}{2M_j} + \lambda K \sum_{j=1}^{4} f_j(s)X(x)R(q_j).$$

The functions X, R are bounded smooth coordinate cutoffs, equal to their arguments on the admitted trajectories. Fixed smooth bumps f_j switch the couplings as a prepared clock moves through them. Pulse widths and observation time $T > 0$ stay fixed. The interaction forces are globally bounded; harmonic forces are uniformly bounded on the fixed receiver energy preparation and bounded apparatus energy class. All kinetic terms are positive.

The reference output $x^0(t)$ determines $z_0 = (x_0, P_0, y_0, Q_0)$: its first four derivative rows have triangular diagonal $1, 1/\mu, g/\mu, g/(\mu\nu)$. Analyticity supplies four independent time evaluation rows. Choosing sufficiently narrow but fixed positive-width bumps preserves independence of the integrated matrix

$$\mathcal{A}_{j\cdot} = K \int_0^T f_j(s_0 + vt) e_x \Phi_t \, dt.$$

For incoming apparatus support widths b in fixed component units, Hamilton's equations and finite-horizon integration give

$$\boldsymbol{\pi}(T) = \boldsymbol{\pi}(0) - \lambda \mathcal{A} z_0 + O(\lambda b + \lambda^3), \qquad \sup_t \|z(t) - \Phi_t z_0\| = O(\lambda b + \lambda^2).$$

These estimates include clock reaction. The clock leaves the pulse supports with positive momentum, after which the four probe momenta are conserved. Exact access to them gives the calibrated estimate $\widehat{z}_0 = -\mathcal{A}^{-1}\boldsymbol{\pi}(T)/\lambda$, with error $O(b/\lambda + b + \lambda^2)$. Choosing $b = \lambda^3$ makes reconstruction and disturbance both $O(\lambda^2)$ at fixed nonzero clock mean energy. Initial canonical position error times maximum canonical momentum disturbance is then $O(L_* P_* \lambda^4)$, in action units for fixed units L_*, P_*.

C068–C069 and the full autonomous proof specify the compact preparation domain, cutoff continuation and uniform estimates; B37 audits the pointer and observability precedents. The result uses increasingly precise preparation, exact final records and known dynamics. All reported cuts may be refined from the same four records after fixed latency T. R07 supplies the following separate new-information test. The upper mass, duration and force resources alone leave the autonomous readout's action product with zero infimum.

12. A delayed observer with an unresolved force

A single-particle information task yields a sharp prediction lower bound. Keep exact records through time zero, including (q_0, p_0), but allow no new record before the prediction time $\ell > 0$. Let the unknown external force be any measurable f with $|f| \le F$. This explicitly differs from the reconstructed known dynamics above. For mass $m > 0$,

$$v := p(\ell) - p_0 = \int_0^\ell f(s)ds, \qquad u := q(\ell) - q_0 - p_0\ell/m = \frac{1}{m}\int_0^\ell (\ell - s) f(s) ds.$$

The forces $+F$ and $-F$ have identical accessible records. Their endpoint separations imply worst-case prediction errors of at least $F\ell^2/(2m)$ and $F\ell$. Inertial prediction attains both, so the product of coordinate minimax errors is exactly $F^2\ell^3/(2m)$, in action units.

The joint region carries more information than these two error bars. Writing $d = v/(F\ell)$ and $z = mu/(F\ell^2)$, it is exactly

$$-1 \le d \le 1, \qquad |z - d/2| \le (1 - d^2)/4.$$

At fixed impulse, maximal early positive force and maximal late negative force give the upper boundary; reversing their order gives the lower one. Convex combinations fill the region. Integrating its width gives canonical area $2F^2\ell^3/(3m)$, without a 2π normalization. This area describes admissible alternatives, not a lower action for each individual motion.

C070–C071: proof, source audit. The area closes with force budget or delay. R08 supplies the cut-composition result below.

13. Exact cuts and observed-position information

At fixed total horizon $T = a + b$, an unobserved cut preserves the bounded-force reachable set exactly. Write $S_b(q,p) = (q + bp/m, p)$ and let K_t be the increment lens above with duration t. Splitting and concatenating admissible force inputs proves

$$K_T = S_b K_a + K_b.$$

The sum is Minkowski addition; every finite unobserved refinement therefore leaves the old endpoint region and its canonical area unchanged. Replacing K_a by its marginal rectangle admits incompatible position–momentum pairs and strictly enlarges the propagated set.

An exact, non-disturbing phase-state record at the cut leaves a translate of K_b, with area $2F^2b^3/(3m)$. An exact position record instead leaves a momentum interval of width

$$D = Fa\,[\sqrt{2-4\zeta} + \sqrt{2+4\zeta} - 2], \qquad \zeta = \frac{m(q_a - q_0 - p_0 a/m)}{Fa^2} \in [-1/2, 1/2].$$

Its terminal area is $2F^2b^3/(3m) + Fb^2D/m$. To obtain the added area, shear the momentum segment vertically: each convex section lengthens by D over a transverse width Fb^2/m. As $b \to 0$, this area vanishes uniformly, but the central-record momentum width tends to $2(\sqrt{2}-1)FT$. Area closure alone therefore does not establish momentum recovery.

C072–C073: proof and assumptions, B39 audit. The finite-precision extension follows.

14. Finite precision and residual delay

A record $|q - y| \le \varepsilon$ clips the cut lens to a convex set C with position interval $[l, r]$. Let its increasing momentum-fibre endpoints be $L(q), U(q)$, its area A_C, its width $w = r - l$, and $D_* = U(r) - L(l)$. The exact terminal set is $S_bC + K_b$, with area

$$|S_bC + K_b| = A_C + \frac{2F^2b^3}{3m} + 2Fbw + \frac{Fb^2}{m}D_*.$$

After the inverse shear, each future-force segment adds a transverse width $w + sD_*/m$; integrating its extrusion gives the two mixed terms. Exact position and unobserved full-lens records recover the preceding formulas. Since $w \le 2\varepsilon$, $D_* \le 2Fa$ and $A_C \le 2Faw$,

$$|S_bC + K_b| \le 4FT\varepsilon + \frac{2F^2ab^2}{m} + \frac{2F^2b^3}{3m} \longrightarrow 0$$

for every joint precision/delay limit at fixed $F, m, T = a + b$. Central records still leave finite momentum uncertainty. R10 therefore tests two position records, separating recovery of both coordinates from area closure. C074–C075: complete formula and proof, B40 audit.

15. Two records and recovery of momentum

Positions recorded at $t_1 = T - b - \delta$ and $t_2 = T - b$, with errors bounded by $\varepsilon_1, \varepsilon_2$, give the momentum estimate $\widehat{p} = m(y_2 - y_1)/\delta$. Define

$$B = \frac{m(\varepsilon_1 + \varepsilon_2)}{\delta} + \frac{F\delta}{2}. \qquad R_p = B + Fb, \qquad R_q = \varepsilon_2 + \frac{bB}{m} + \frac{Fb^2}{2m}.$$

These errors of $\widehat{p}$ and $\widehat{q}_T = y_2 + b\widehat{p}/m$ are simultaneously attained by constant force and adverse record errors. The compatible-set area is at most $4R_qR_p$. If $\delta, b \to 0$ and $(\varepsilon_1 + \varepsilon_2)/\delta \to 0$, both coordinates are recovered.

For equal errors, the established finite-difference optimum is $\delta_* = 2\sqrt{m\varepsilon/F}$. Choosing positive delay $b = \delta_*$ gives $R_p = 4\sqrt{mF\varepsilon}$, $R_q = 7\varepsilon$, and an action-error product $28\sqrt{mF}\varepsilon^{3/2} \to 0$. R11 tests minimax lower bounds. C076–C077: proof, exact prior-art match.

16. A gap that survives dense noisy records

At fixed position tolerance $\varepsilon > 0$, even the entire noisy position history leaves a sharp momentum uncertainty. Put $d = 2\sqrt{m\varepsilon/F}$, $t_2 = T - b$, $t_1 = t_2 - d$, and assume $t_2 \geq 2d$. Starting from zero displacement and velocity, forces $-F, +F$ for successive times $d/2$ prepare displacement $-\varepsilon$ with zero velocity at t_1. Force $+F$ for the next d reaches $+\varepsilon$ and momentum Fd. The entire observed displacement stays in $[-\varepsilon, \varepsilon]$.

This trajectory and its opposite have the same initial state and admit the same inertial position record, with opposite allowed record errors. Continuing their forces through the blind interval yields terminal half-separations

$$P_* = Fd + Fb, \qquad Q_* = \varepsilon + \frac{Fdb}{m} + \frac{Fb^2}{2m}.$$

Every estimator has worst-case coordinate errors at least P_*, Q_*. The preceding two-record estimator attains both; these are therefore exact coordinate minimax risks for two records and for the complete noisy history. At zero blind delay their product is $2\sqrt{mF}\varepsilon^{3/2} > 0$. Densifying records preserves this bound; improving their tolerance closes it.

The construction is established in Seeber–Haimovich Proposition 3.1. R11 connects it to reachable fibres and delayed phase prediction. The general principle is that, for symmetric convex inputs and bounded linear record errors, the central compatible fibre maximizes scalar output uncertainty: any compatible pair has an admissible half-difference, and the symmetric central pair realizes its diameter.

C078–C079: proof and precise information classes, B42 audit. The following composition test settles the proposed transfer to mass universality.

17. Worst-case composition and the cost of aggregate records

Product bounded-force/error classes add scalar minimax radii linearly. For positive masses with known initial states, complete position records of errors $\varepsilon_i > 0$, forces $|f_i| \leq F_i$ and common terminal time $T \geq \max_i 4\sqrt{m_i\varepsilon_i/F_i}$, put $M = \sum_i m_i$, $E = \sum_i m_i\varepsilon_i/M$ and $F_\Sigma = \sum_i F_i$. With all constituent records retained, the exact centre position and total momentum risks are

$$Q_A = E, \qquad P_A = 2\sum_i \sqrt{m_i F_i \varepsilon_i}.$$

The central compatible fibre is a Cartesian product. Its support in a scalar sum direction is the sum of the supports, attained by aligned symmetric extremizers; R11's midpoint argument converts that radius into exact minimax risk. For n identical constituents the product is therefore $H_A = nH_1$. It does not obey A08's invariant variance-coefficient law.

Keeping only the weighted sum of the records gives exactly the single-body force/error class (M, F_Σ, E). To lift any aggregate force f and error e back to the original constituents, choose $f_i = F_i f/F_\Sigma$ and $e_i = \varepsilon_i e/E$, preserving each initial state. Thus its exact risks are

$$Q_B = E, \qquad P_B = 2\sqrt{MF_\Sigma E} \geq P_A.$$

Equality holds precisely when $F_i/(m_i\varepsilon_i)$ is common to all constituents. Otherwise losing the individual records strictly increases the momentum risk in the same mechanical model.

If mass-only nonnegative coordinate radii are closed under product composition for every positive mass, additivity forces $Q(m) = q_*$, $P(m) = p_*m$. Their action product is extensive; a mass-independent product must be zero. Fixed acceleration bound $F(m) = am$ and common precision realize the positive extensive case, whose value still comes from a and the record band. R13 next examines the finite-horizon regime before the hidden pair can be prepared.

C080–C081: complete proof and information classes, B43 audit. These products multiply coordinate minimax risks and are not simultaneous error lower bounds on each motion.